



Figure 2: Flow of processes in a blockchain environment

programming language, the consensus protocol, and the platform's popularity and activity. These factors need to be analysed so that a high performance blockchain application can be developed that is compatible and has a high throughput.

Let's look at the key features integrated in the prominent platforms for blockchain programming and smart contracts.

Eris (<https://erisindustries.com/>): This is a powerful and multi-featured platform for blockchain programming. Smart contracts can be written and executed on Eris effectively. The private blockchain networks can be programmed with a higher degree of privacy and security for multiple applications. This platform is popular among blockchain programmers for the automation of smart contracts. The programmed smart contracts are used to analyse and validate the transactions in a secured environment.

HydraChain (<https://github.com/HydraChain/hydrachain>): HydraChain is used for permissioned distributed ledgers. The key feature of HydraChain is that it is fully compatible with Ethereum Protocol. The primary domain of HydraChain includes consortiums or private chain based setups. In addition, native contracts that can be easily customised and deployed can be programmed using HydraChain.

OpenChain (<https://www.openchain.org>): OpenChain refers to the free and open source distributed ledger platform. It is used for high-performance scalable applications for organisation and enterprise deployments. The execution speed is much faster than other platforms.

OpenChain integrates the client-server architecture, in which there is more reliability and consistency as compared to the traditional peer-to-peer architecture. There is support for multiple signatures in OpenChain which provides multiple keys for authentication and validation.

The key features of OpenChain include:

- No mining charges
- Security using digital signatures
- Multiple layers of control
- Instant validation of transactions
- Immutable
- Transactions audit
- Transparency and validation control

Ethereum (<https://www.ethereum.org>): Ethereum is a free and open source platform for blockchain programming. It can be used for writing and deploying any type of decentralised applications, and can be used on multiple platforms with high security levels and flexibility. Smart contracts can be programmed and deployed using Ethereum. This platform is generally used for public blockchains. The support programming languages in Ethereum include Go, Python and C++.

Hyperledger (<https://www.hyperledger.org>): This is another open source platform for advanced applications of blockchain technology. It is used widely for enterprise applications and for working with private blockchain networks. The Python programming language is used for working with Hyperledger, which is integrating a number of projects for blockchain programming with distributed ledgers. These projects include Hyperledger Besu, Hyperledger Burrow, Hyperledger Fabric, Hyperledger Indy, Hyperledger

Iroha and Hyperledger Sawtooth. There is a rich set of tools in Hyperledger including Avalon, Caliper, Cello and Explorer. These tools are used for monitoring and programming with the assorted dimensions of the blockchain network under development.

Scope for implementation in real-world Web and mobile applications

To program real-world applications with blockchain technology, there are many programming languages that are used including Python, JavaScript, Java, PHP, Go, etc.

Programming with blockchain uses the base of cryptography hash functions which are required to encrypt the blocks and the transactions. In addition, programming the smart contracts also integrates the cryptography libraries so that the overall communication and data processing in the blockchain are fully secured.

The following scripts form the foundations of blockchain programming for Web or mobile applications in multiple domains, for which the transactions must to be implemented in secured scenarios.

Block creation and activation: Shown below is the code snippet for writing a block in the blockchain environment for Web or mobile applications in which multiple transactions or processes are expected.

```
const SHA256 = require("crypto-js/sha256");
class SecuredBlock
{
  constructor(BlockIndex, CurrentTime, DataBlock, PreviousHashValue = '')
  {
```